

A key research topic in the Biosecurity Research and Analysis Branch is understanding the geographic distribution of established vertebrate pest animals and weeds, their economic impacts and their direct and indirect non-market environmental and social impacts. Much of the foundational work in this area was funded by the Australian Government's Established Pest Animal and Weed Management Pipeline Program and the Supporting Communities Manage Pest Animals and Weeds Program.

ABARES Invasive Species Program has an ongoing body of work to map the national distributions (and, where possible, the relative abundance) of key vertebrate pest and weed species in Australia. This work is being conducted in collaboration with jurisdictions and other stakeholders, with the guidance of a project steering committee.

The following spatial datasets have been released to date:

- [Current National Relative Abundance of Fallow Deer in Australia \(2026\)](#)
- [Current National Relative Abundance of Sambar Deer in Australia \(2026\)](#)
- [Current National Occurrence of Feral Deer in Australia \(2026\)](#)
- [National Feral Pig Current Distribution in Australia \(2024\)](#)

While the statistical methods are similar, the relative abundance scales cannot be compared between the species (fallow deer, sambar deer and feral pigs).

Project background

This work originated in the National Vertebrate Pests and Weeds Distribution project, which ran from 2019 – 2025. The project identified what is required to collate, build and maintain national distribution datasets for invasive species in Australia and addressed fundamental data gaps and challenges in creating nationally consistent, up-to-date maps of pest animals and weeds distributions for Australia. These include:

- Difficulties in making meaningful comparisons across state and territory or regional boundaries when integrating disparate datasets, due to invasive species data being collected using different methods and sampling protocols at varying geographic scales and time periods.
- Geographic bias in how field-based data for widespread invasive species are collected, with the majority of these biased towards either control work targeted to areas of known occurrence, or towards areas of higher human population (e.g. citizen science-based observations or jurisdictional-based atlases).

The project included a collaboration with CSIRO to develop and apply analytical modelling methods for presenting a reliable picture of national pest animal and weed distributions from disparate data sources. These modelling methods were published in 2024 in an R package and associated journal article: [‘RISDM’: species distribution modelling from multiple data sources in R](#).

Vertebrate pest animal and weed species cause significant impacts on industry and significant resources are spent trying to reduce these impacts. Past studies have quantified impacts and expenditure. In November 2023, ABARES released the [Cost of established pest animals and weeds to Australian](#)

agricultural producers. This body of work updated the estimates by drawing on multiple sources of information. Estimates were provided both nationally and for states and territories, and for broad agricultural industries.

Many pest animals and weeds cause significant impacts (both positive and negative) on the well-being of people through their direct and indirect effects on human health, biodiversity, society and ecosystem services and function. These impacts on society and the environment are not reflected in many of the observable market outcomes of the economy and cannot be measured in the same way as the production and cost effects, making it difficult to put a “price tag” on all the impacts that invasive species have in the community.

ABARES has released several pieces of research in this area:

Source: <https://www.agriculture.gov.au/abares/research-topics/invasive-species/distribution-and-impacts>